

INCIDENT REPORT 6

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Report ID: IR-006
Incident Name: Privilege Escalation — Linux Sudo Abuse
Date/Time: May 5, 2026, around 12:50–12:54
Severity: HIGH
Status: Resolved (lab environment)
Environment: Wazuh 4.7.5 Mini SOC Home Lab
Analyst: Jazz

1. EXECUTIVE SUMMARY

At approximately May 5, 2026 around 12:50–12:54, Wazuh SIEM detected sudo-related privilege escalation behavior on the Ubuntu endpoint. The activity included successful sudo execution as root and multiple failed sudo password attempts.

The user account jazz executed commands requiring root privileges, including access to /root and /etc/shadow. The lab also simulated failed sudo authentication by entering an incorrect password three times.

This activity simulates a common post-exploitation step where an attacker attempts to elevate privileges after gaining initial access. The incident maps to MITRE ATT&CK T1548.003 — Sudo and Sudo Caching.

2. TIMELINE

May 5, 2026 around 12:50:11

Wazuh detected successful sudo activity on ubuntu-endpoint.

May 5, 2026 around 12:50:43

Wazuh generated “Three failed attempts to run sudo.”

May 5, 2026 around 12:51

Additional sudo session opened and closed events were logged.

May 5, 2026 around 12:54

Further sudo activity appeared in Wazuh Security Events.

May 5, 2026 after alert review

The analyst reviewed the Wazuh alert details and auth.log evidence.

May 5, 2026 after investigation

The event was confirmed as a true positive lab simulation.

3. DETECTION SOURCE

Tool: Wazuh SIEM/XDR
Wazuh Version: 4.7.5
Detection Area: Linux privilege escalation monitoring
Log Source: Linux /var/log/auth.log and Wazuh Security Events
Affected Agent: ubuntu-endpoint
Agent OS: Ubuntu 22.04.5 LTS
User Account: jazz
Target Account: root

Observed Wazuh Rule IDs:

- 5402 — Successful sudo to ROOT executed
- 5404 — Three failed attempts to run sudo
- 5501 — PAM: Login session opened
- 5502 — PAM: Login session closed
- 5503 — PAM: User login failed

Severity:

- Rule 5402: Level 3
- Rule 5404: Level 10
- Rule 5503: Level 5

4. MITRE ATT&CK MAPPING

Tactic: Privilege Escalation / Defense Evasion
Technique: Abuse Elevation Control Mechanism
Sub-technique: T1548.003 — Sudo and Sudo Caching

Explanation:

The activity maps to Privilege Escalation because sudo allows a normal user to run commands as root. Attackers often abuse sudo access after gaining a low-privilege shell to take full control of a Linux system.

Potential attacker next steps if successful:

- Read sensitive files such as /etc/shadow
- Modify system files
- Add users or SSH keys

- Install persistence
- Disable logging
- Run tools as root

5. INVESTIGATION FINDINGS

Source:

Local activity from user jazz on ubuntu-endpoint.

Target:

Root account and root-only system resources.

Commands observed:

- sudo ls /root
- sudo cat /etc/shadow
- sudo su root

Observed behavior:

- User jazz executed sudo commands.
- User jazz accessed root-only resources.
- The lab simulated failed sudo authentication.
- Wazuh detected successful sudo execution.
- Wazuh detected three failed sudo attempts.
- The alert appeared in Security Events.
- The activity mapped to MITRE T1548.003.

Evidence reviewed:

- Wazuh Security Events showing Rule 5402
- Wazuh Security Events showing Rule 5404
- Ubuntu /var/log/auth.log evidence
- Terminal evidence showing sudo commands
- MITRE ATT&CK Sudo and Sudo Caching dashboard

Compromise confirmed:

No. This was a controlled lab simulation.

Unauthorized access confirmed:

No production unauthorized access occurred.

6. RISK ASSESSMENT

Risk: HIGH

Reason:

Sudo activity becomes high risk when it involves failed authentication attempts, access to sensitive files, or attempts to become root. If an attacker abuses sudo successfully, they may gain full control of the Linux endpoint.

The risk is especially high if:

- The user should not have sudo access
- /etc/shadow is accessed
- Multiple failed sudo attempts occur
- Sudo occurs after suspicious SSH login activity
- The user opens a root shell

In this lab, the risk was controlled because the activity was intentional.

7. VERDICT

Verdict: TRUE POSITIVE — Lab Simulation

The alert is a true positive because the sudo activity actually occurred and Wazuh detected it. It was not a false positive because the logs clearly show successful sudo usage and failed sudo authentication.

In production, this event would require investigation to confirm whether the user was authorized.

8. RESPONSE ACTIONS TAKEN

1. Reviewed Wazuh Security Events.
2. Identified sudo-related alerts.
3. Verified Rule IDs 5402 and 5404.
4. Verified the affected agent as ubuntu-endpoint.
5. Reviewed /var/log/auth.log evidence.
6. Confirmed the user account was jazz.
7. Confirmed the target account was root.
8. Confirmed MITRE mapping to T1548.003.
9. Documented the incident.

9. RECOMMENDATIONS

1. Review sudo access regularly.
2. Limit sudo privileges using least privilege.
3. Monitor sensitive sudo commands such as:
 - sudo cat /etc/shadow
 - sudo su
 - sudo /bin/bash
4. Alert on multiple failed sudo attempts.
5. Review /var/log/auth.log regularly.
6. Check /etc/sudoers and sudo group membership.
7. Investigate sudo activity after suspicious SSH logins.
8. Lock accounts involved in unauthorized sudo attempts.
9. Isolate the host if root compromise is suspected.
10. Review persistence locations such as cron jobs, systemd services, and SSH keys.

10. FINAL ANALYST NOTE

This incident demonstrates Wazuh's ability to detect Linux privilege escalation behavior. The lab confirmed that Wazuh can collect sudo/authentication logs, generate alerts, map activity to MITRE ATT&CK, and support SOC-style triage for privilege escalation events.